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XML Xtreme! – Learn How to Query XML in SQL Server

SQLRV, LLC
HEALTHCARE INFORMATICS
https://www.sqlrv.com

Just who is this guy?

Principal Data Architect, Population Health

Adjunct Instructor

Ambassador Programs

Speaker – various events

B.S. Computer Science

M.S. Computer Information Systems

M.S. Health Informatics

Doctor of Healthcare Administration

40 Years

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Something to consider…

"However difficult life may seem, there is always something you can do and succeed at."

-- Professor Stephen Hawking, 1942-2018

Theoretical Physicist

Agenda

- Some background info on XML
- Introduction to use case
- Opening XML documents for queries
- Querying simple structures
- Joining with SQL Server tables
- Querying complex structures
- Processing nested structures
- Query optimization
- In closing…

Some background info

- XML (Extensible Markup Language)
  - Developed from previous SGML (Standard Generalized Markup Language)
  - Accepted as standard in 1998
- Initial document structure enforced by DTD (Document Type Definition)
  - Not expressed in XML
  - Somewhat limited
- XSD (XML Schema Definition) v1.0 in 2001
  - Expressed in XML
  - Adds Namespaces
- XSD v1.1 released in 2004
  - Fixed some flaws and added some new features
  - Still not widely adopted (ex: MS .NET only uses v1.0)

Some background info

- What is XML?
  - Text-based markup language often compared with HTML (Hyper-Text Markup Language)
  - Identify data using tags (identifiers enclosed in angle brackets, like this: <...>)
  - Unlike HTML, XML tags tell you what the data means, rather than how to display it
  - Its structure is typically constrained by a schema definition file (XSD) which is also written in XML
- Isn't XML complex?
  - XML is based upon simple rules for flexibility
  - The power of XML comes from various combinations

Introduction to Use Case

- Share patient data between disparate systems
  - Allow hospitals and doctors to send patient data without regard to their systems
  - Allow requests for information for a specific patient
- Requirements:
  - Message file may have multiple recipients (CC)
  - Multiple patients may be presented
  - Must handle various types of data
    - Visits/Encounters
    - Diagnoses/Procedures/Laboratory/Results
  - Will use standard vocabulary systems (CPT, LOINC, etc.)
- This solution will be greatly simplified as real-world cases are much more complex!

Patient Message Schema

- Schema Components
  - Elements – The main component within XML
  - Attributes – Characteristics of an element
- Data Types
  - Built-in – Basic types supplied by XSD
  - Simple Types – Used to identify a single attribute
  - Complex Types – Used to identify elements – may have nested elements in the definition
  - Extensions – Extend Complex types with additional attributes
- Groupings
  - Sequence – Allows a series of elements
  - Choice – Allows a choice of one of a series

Patient Message Schema

XML File Structure

- Schema validates:
  - File format and structure
    - Element hierarchy
    - Sequences & choices
    - Occurrence validation
    - Optional and required attributes
  - Some content validations
    - Attribute enumerations
    - Data types and formats (where applicable)
- XML file size argument
  - Verbose requires full tags and descriptive names
  - Concise versions tend to be difficult to read

Patient Message Example

Reading from XML data type

- Using .nodes() method
  - Used to shred XML data type instances into relational data
  - Allows to identify nodes that will be mapped into a new row
  - Syntax: nodes (XQuery) as Table(Column)
- Using .value() method
  - Extract value from an XML instance
  - Typically used to retrieve column data
  - Syntax: value (XQuery, SQLType)
- Uses a subset of XQuery expression syntax
  - Used to traverse XML trees
  - Specifies access to elements and attributes
  - Can perform complex queries

Patient Message Example

Query Optimization

- XML is NOT set-based!
  - XML functions process for each row of output
  - Behaves much in the same way as UDFs
  - It may look set-based, but it’s not
- To minimize the impact:
  - Extract data to a table (regular, temp, variable) to encourage set-based operations
  - Filter where possible using XQuery instead of the WHERE clause

XQuery

- Complete language in of itself
  - References within XML
  - Performs conditional logic, loops, variables, etc.
  - Compact syntax yet easily understood
- Not used by itself
  - Not a separate, stand-alone language
  - Used within SQL Server & other database tools
  - Used in conjunction with other development languages

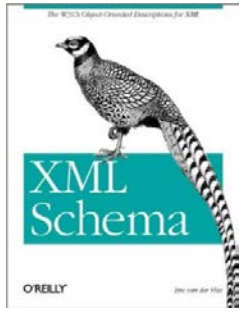
Would you like to know more?

No! No more! Please! **Please!**

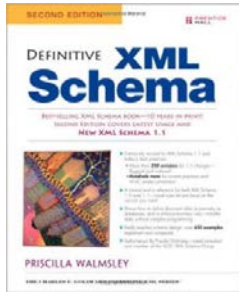
In closing…

- Barely scratched the surface
  - XQuery is rather extensive – a language of its own
  - Complex XSD implementations are virtually limitless
  - Additional presentation on XSD capabilities
- Simple examples are shown here
- Much more complex schemas are typical
  - Recursive definitions
  - Multiple XSD files for a single schema
- This case model (XSD & SQL) can be extended

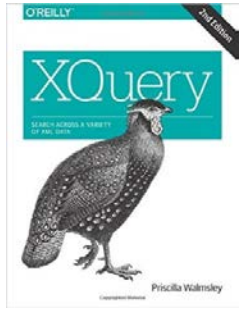
Useful resources



O'Reilly *XML Schema* by Eric van der Vlist



Prentice Hall *Definitive XML Schema, 2nd Edition* by Priscilla Walmsley



O'Reilly *XQuery, 2nd Edition* by Priscilla Walmsley

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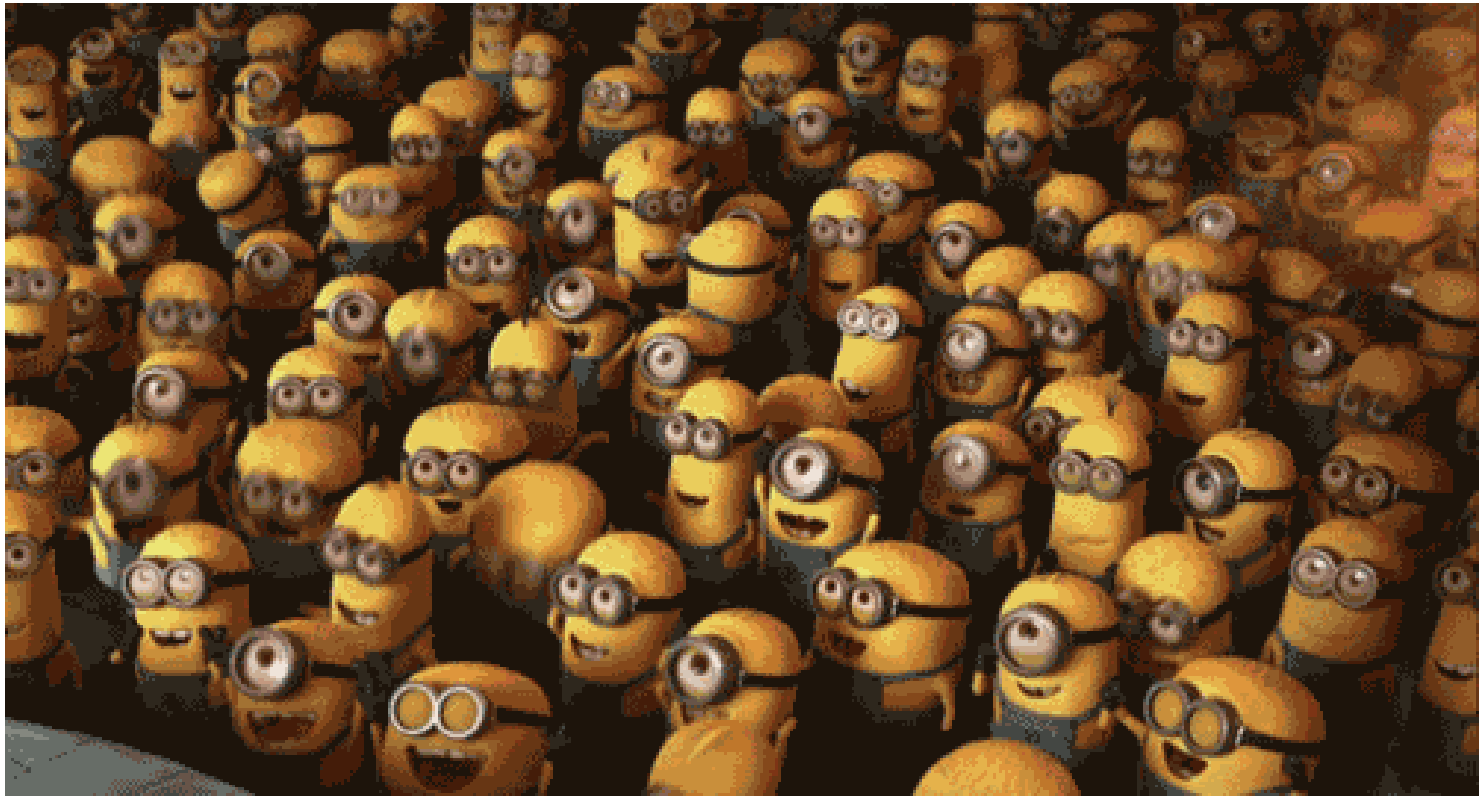

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